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25.10.2022

IC-Quotation

Project: Hyundai-Wia
Friction Welding Machine KUKA Genius *plus*
Budget Indication no.: 20012106.0/AG

Dear Mr. Hwang,

We thank you for your inquiry for a Friction Welding Machine for Tulips and Drive Shafts. In response, please find enclosed our Quotation for a Friction Welding Machine, Type KUKA Genius *plus*.

We hope our budget - indication meets your requirements. Should you have any questions or require further information, please do not hesitate to contact us.

Yours sincerely,

KUKA Deutschland GmbH
Advanced Welding Solution

Andreas Lenz
Global VP Advanced Welding Solutions

Andreas Gistl
Director Sales & Business Development Asia & Pacific /BU AWS

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1 CONFIDENTIALITY OBLIGATION

This indication contains technical, commercial, and organizational information and data **(hereinafter: "confidential information")** which are confidential and the property of KUKA Deutschland GmbH **(hereinafter: "KUKA Deutschland")**. The parties are obligated to treat the confidential information, made available to them by the other party, either before or after placement of this order, strictly confidential, not to disclose it to third parties or use it for any other commercial purposes. The confidential information contained in this indication has been disclosed solely for the purpose of assessment and shall only be disclosed to such persons as are involved in the indication process.

2 BASICS

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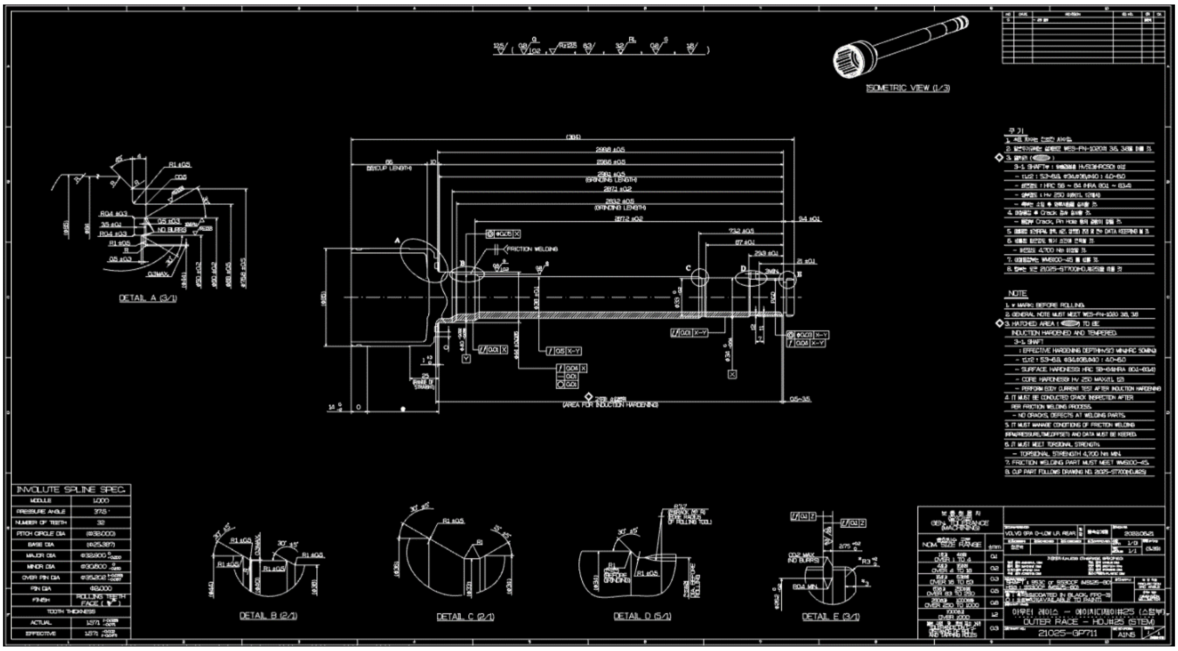
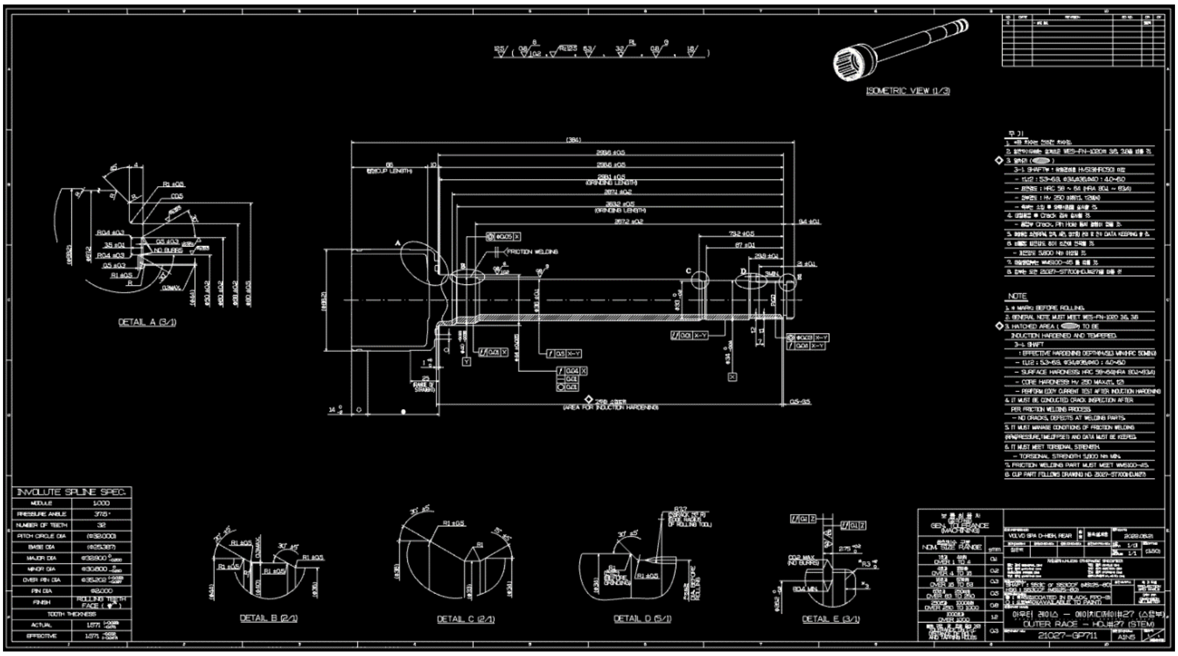
3 DELIVERED DOCUMENTS:

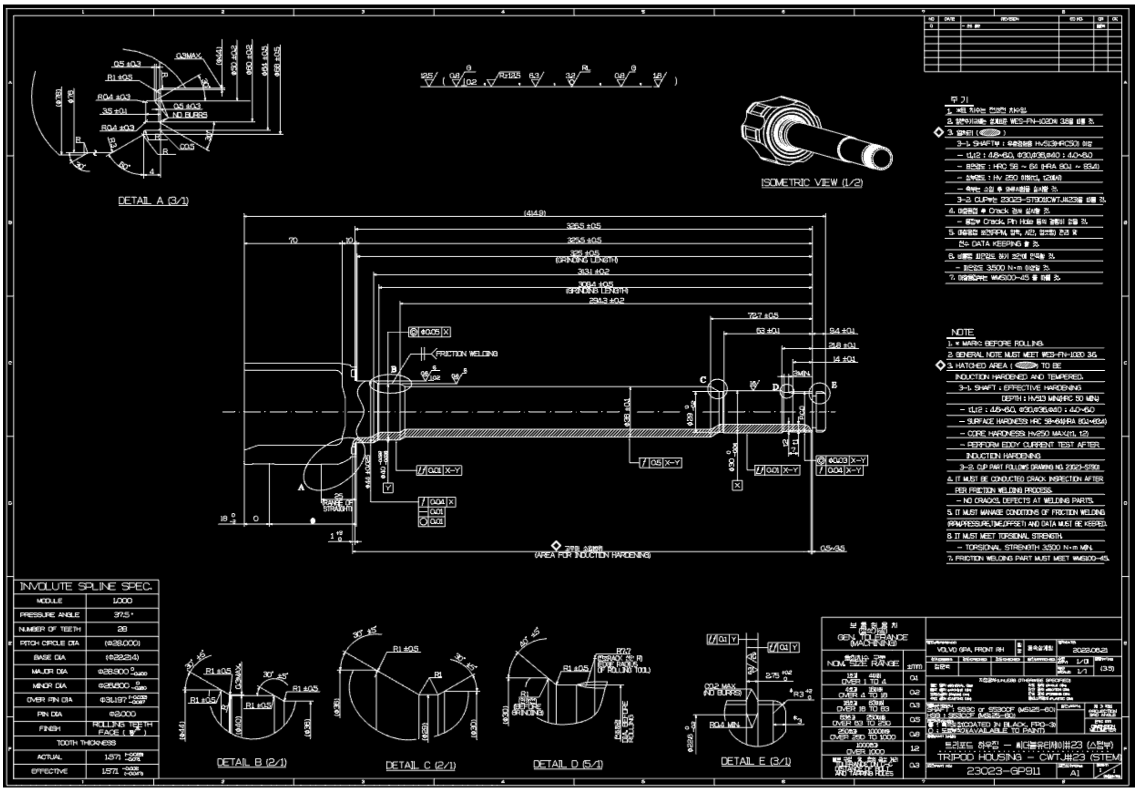
Volvo GPA 21025-GP711



Volvo GPA 21027-GP711



Volvo GPA 23023-GP911



- for reference only-

4 SCOPE OF SUPPLY OF KUKA DEUTSCHLAND

4.1 Description

4.1.1 Basic machine - KUKA Genius *plus®*

KUKA Friction Welding Machines are designed and built to incorporate the principles of "Combined Friction Welding" (EN ISO 15620) - i.e. the parameters can be tailored exactly to suit the welding application and quality requirements.
The machine is built to a modular design. The hydraulic system and the electrical control are integrated in an ergonomic and space-saving manner in the machine frame including the following components:

- Genius spindle drive
- Genius software package
- Machine bed
- Headstock
- Hydraulic system (Rexroth) with automatic temperature control
- Numerically controlled process feed motion (HNC)
- Incremental length measuring system
- Machine control with process and parameter monitoring
- Access management
- Touch Panel with additional stainless-steel keypad and trackball
- Integrated electrical and control cabinet
- Remote Service Client for remote diagnostics
- Drawing of oil drip pan
- Safety devices

The following description provides more detail to each of the items.

Spindle Drive Genius *plus*®
Servomotor (90kW) protected by thermal overrides. Direct drive concept with reduced inertia and low-vibration, bend free transmission of drive power to the spindle.
During the welding process, the machine control records the exact speed and regulates the drive within the preset range of index values for speed in real time (or optionally in a graphical representation of the preselected setpoints).

Software Package Genius *plus*®
The drive technology with the hardware and software package plus was developed by KUKA specifically for standard applications an includes, inter alia, monitoring of graphical envelope curves, an oscilloscope function for actual values for the purpose of documentation and fault analysis and matched hydraulic valves. Adaptation and integration into the machine control are executed by means of a special interface.

Machine Bed
Machine bed with high degree of inherent rigidity, without bending moment. Simplified installation without anchoring of the machine bed to the floor.

Headstock
Cast structure, optimized using the Finite Element Method for an extended speed range up to 3,000 min^{-1}.

Hydraulic System (Rexroth) with Automatic Temperature Control
Electronically controlled hydraulic power pack of compact design. Reduction in energy consumption due to automatic adjustment of pump capacity.
Arrangement of the hydraulic valves in close vicinity to the electrical loads allow short reaction times.
To ensure a reproducible process sequence, it is necessary that the temperature of the oil in the hydraulic system is within a clearly defined range.
In order to have constant conditions at the start of production, the oil is brought to operating temperature and held at the target temperature during production by means of a cooling system.

Sensor System
The sensors are positioned centrally and clearly arranged. All changeable sensor parameter sets are set centrally via IO-link technology, stored and displayed in the control panel. No additional adjustment and reading of the values at the sensor itself is required, which simplifies maintenance and servicing procedures.

Numerically Controlled Process Axis

Power is transmitted through a closed flow of force between the spindle, feed cylinder and component back stop to the center-line of the component.
By means of the numerically controlled process sequence, position, speed and pressure are regulated, thus promoting a reproducible machine performance which is independent of temperature and friction.

Advantages:

- Very simple parameterization of slide stroke
- No manual resetting of the slide control is required when retooling the machine for another type of component
- All settings relating to the slide control are contained in the data sets of the welding program, together with all other welding parameters, and consequently are automatically documented and archived
- Precise and smooth slide stroke

Incremental Length Measuring System

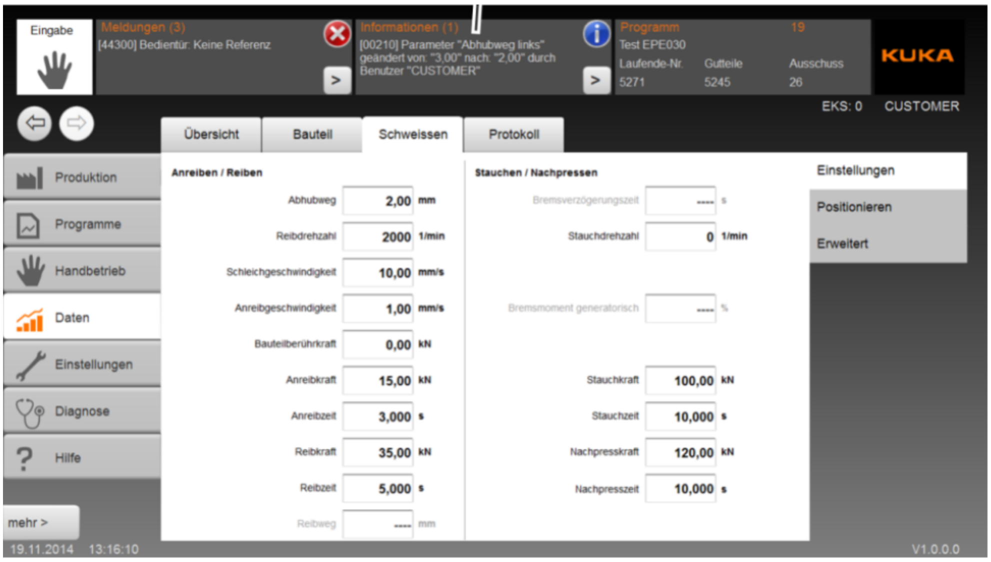
Thanks to its high degree of positioning accuracy, the incremental length measuring system makes referencing the slide position and exact registration of each slide movement easy. The length measuring system is resistant to contamination due to its casing and special scanning principles.

Machine Control with Process and Parameter Monitoring

Central unit for controlling machine and process, monitoring welding parameters and documenting production.
The machine control and monitoring concept developed by KUKA combines the control and monitoring functions in one standardized system, thereby reducing the number of interfaces required and facilitating machine operation. The visualization is menu-driven by means of a multi-touch panel.
The KUKA computer technology (Quad-Core) controls the welding process accurately to the millisecond and can react in real time to changing process conditions.

Setting Machine and Process Parameters

Machine settings and process parameters can be stored using component reference numbers/codes. The use of a SQL server enables up to a 1000 sets of data to be saved.



- for reference only-

All the important parameters which influence the welding process can be monitored. For this purpose, the following tolerance limits can be set as required:

- max. and min. common relative initial length, so-called initial position of both components, in 1/100 mm
- max. and min. spindle speed during the friction phase in 1/min
- max. and min. friction pressure during the friction phase in bar
- max. and min. friction displacement (friction shortening) at the end of the friction phase in 1/100 mm
- max. and min. time for the friction phase in 1/1000 s
- max. and min. forging pressure during the forge phase in bar
- max. and min. holding pressure during the holding phase in bar
- max. and min. total shortening after welding in 1/100 mm
- Monitoring of braking operation (time and average pressure)
- When used, max. and min. angular position of the spindle after positioned welding
- max. and min. common relative final length, so-called final position of both components, in 1/100 mm
- Special function for monitoring extremely short welds (mean value of pressures employed in each welding phase)

Any failure to keep within tolerance limits will be shown on the display. The machine will be locked, and a clear visual signal given. It is not possible to remove the reject weld manually until an authorized person has acknowledged the fault. Further operations such as destroying or marking the component are also possible and can be quoted on request.

Access Management “Euchner EKS light”

The control provides an electronic access management system with identification chip EKS light. Various user levels and access authorizations are available, e.g. machine operator, machine setter, maintenance personnel and KUKA service personnel. The equipment is of type Euchner EKS light FSA modular.

The identification chips are available in five different colors and can be individually coded with 15 EKS levels. The machine is delivered with the following 4 access levels predefined:

Operator:	Operating of machine
Shift leader:	Clearing and monitoring of device fault
Engineer:	Parameter input and setup machine
Maintenance:	Customer setup, mastering, Customer admin

Identification chip	Rights	EKS level
	Operator	2
	Clear monitoring device fault	4
	Parameter input Setup mode	6 8
	Customer setup, mastering, setup pressure for sec-ondary functions, etc. “Customer” admin	10 12
	“KUKA” eadmin	15

The user can also custom-define the identification chips and individually define their EKS levels. No personalization of the chips. All chips of one color have the same EKS level.

- for reference only-

Touch Panel with additional Stainless-Steel Keyboard

The Touch Panel is integrated in the machine housing in an ergonomic and space-saving manner. Thanks to a swivel function, the operator can always set the panel to perfectly suit his working position. The robust heavy-duty design also allows operation when wearing safety gloves. All sets of variable sensor parameters will be displayed centrally on the Touch Panel by means of IO Link technology.

- for reference only-

The robust and indestructible stainless-steel keyboard with integrated trackball has a surface good to touch and ensures perfect ease of operation.

- for reference only-

4.2 Component-specific Tooling (included)

Spindle side

- 1 off 3-jaw universal clamping chuck
- 1 off Inserts for tulip clamping (21025-GP711)
- 1 off Inserts for tulip clamping (21027-GP711)
- 1 off Inserts for tulip clamping (23023-GP711)
- 1 set Jaws for end piece clamping (21025-GP711)
- 1 off Jaws for tulip clamping (21027-GP711)
- 1 off Jaws for tulip clamping (23023-GP711)
- 1 off Inserts for end piece (21025-GP711)
- 1 off Inserts for end piece (21027-GP711)
- 1 off Inserts for end piece (23023-GP711)
- 1 off Backstop for chuck

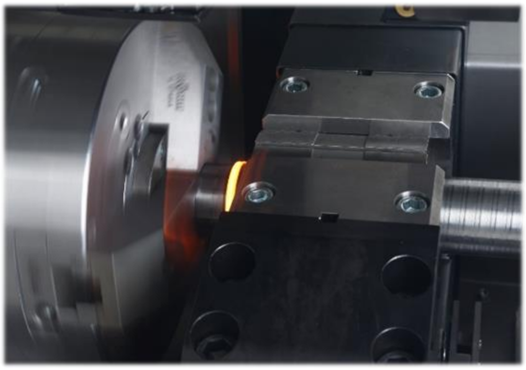
Slide side

- 1 sets Base jaws for self-centering clamping block
- 1 sets Clamping inserts (half shells) for centering clamp
- 1 off Backstop on slide side

- Commissioning and acceptance of 3 acceptance components 21025-GP711 / 21027-GP711 / 23023-GP711)

Spindle Chuck

A collet or three-jaw chuck is provided for clamping the components. By means of adjusting the hardened jaws any diameter within the clamping range may be set. Actuation of the chuck is effected by a rotating hydraulic cylinder connected to a tierod which is, in turn, connected to the chuck via the spindle.



- for reference only-

Clamping Jaws

Component-specific prismatic jaws or half-shells will be provided to clamp components up to 100 mm in diameter on the slide side. These will be inserted into the self-centering clamping block.

Integrated Power Supply and Control Cabinet

The power supply and control cabinet is fully integrated within the machine housing. The lack of external control cabinets and cable ducts serves to save space and means less time and lower costs for transport and installation.

- for reference only-

KUKA RemoteService Client

Integrated KUKA industrial router, to enable remote access to the machine for the purpose of remote maintenance.

The Router RemoteService Client is integrated in the machine/system to get a remote access to subscribers with IP address. Connecting a free Internet connection (minimum required DSL) directly to the router, same will be activated on the Customer's side. This allows KUKA Deutschland to analyze the data of the machine by active release and to introduce targeted remedy measures at short notice.

KUKA Deutschland will provide the equipment and the KUKA Remote Service free of charge during the warranty period in the scope of the warranty obligation with the following advantages:

- Avoidance of time and information loss.
- The RemoteService thus serves as tool for safeguarding your system availability.

Safety Devices

KUKA friction welding machines are built in accordance with the valid accident prevention regulations. Where applicable and achievable for us, VDE guidelines and VDMA recommendations have been taken into consideration.

Oil Drip Pan

The stainless-steel oil drip pan is integrated into the machine design. It has a reservoir capacity for the total amount of oil in the machine.

4.3 Cycle time estimation

At this state of the project KUKA is willing to provide the Machine time for 1 welding procedure as an example:
Time from "push start button" until machine is "ready for unloading".

*Estimated welding process time (component contact up to forging end) f*or ø36 mm (fork to inner tube): *approx. 8-10s*

Estimated machine time without flash removal:	approx. 16,9s – 18,9s
Estimated machine time with flash removal:	approx. 30,4s – 32,9s

Cycle times depend strongly on the process parameters used, therefore those parameters need to be agreed upon first. This can be achieved by either providing proof of the feasibility by Hyundai or weld trials performed during the commissioning phase. As a result, KUKA neither accepts the specified cycle times as performance data, nor the relevance for acceptance procedures.

4.4 Change over time

Change-over time with automatic backstop (for different length up to 500 mm without mechanical changing of backstop) < 1 minute.
Change-over of component specific tooling as described above, depend on worker skills and human factors. Nevertheless, the design of the component specific tooling has been focused on the functionality, usability and lifetime. Change-over should typically < 10 minutes.

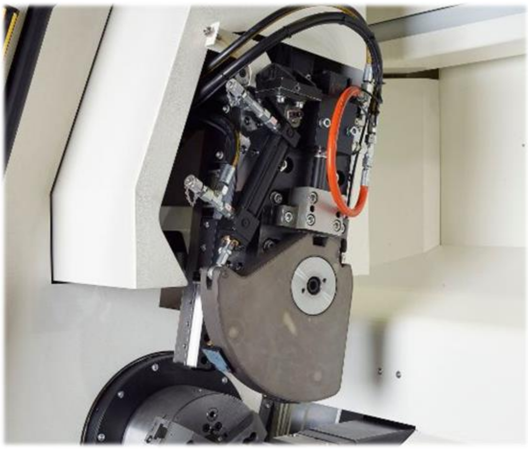
4.5 Equipment Packages (not scope of basis machine)

4.5.1 Flash Removal Package (Option)

- Turning unit
- Tool changer with 1 off cutting heads
- 1 set Turning inserts
- Chip conveyor
- Steady rest

Turning Unit

Turning unit for automatic removal of weld flash and for the execution of various turning operations such as facing the faying/contact surface of the component.
The housing provides protection from flying chips. The utilization of electrical drives and permanently lubricated profiled guides ensures stable progress of the turning operation and reduces the maintenance requirement.



- for reference only-

4.5.2 Tool Changer – Dual (Option)

The turning unit is equipped with an automatic tool changer for two dies. This allows facing of the component's faying/contact surface as well as flash removal. This is necessary e.g. when welding specific material combinations such as aluminum and copper.

4.5.3 Steady Rest (Option)

The purpose of the steady rest is to support and center long components. Thanks to the precise guidance vibration of the component is prevented thus ensuring dimensionally accurate and clean removal of the flash during turning. The steady rest will set automatically to suit the diameter of the component to be turned (up to 100 mm).

- for reference only-

4.5.4 Chip Conveyor (Option)

The chip conveyor integrated in the machine frame can be located to the rear or to the right side of the Genius machine, depending on the space available and customer requirements.

- for reference only-

4.5.5 Cycle Time Reduction Package (Option)

- Automatic opening and closing of operator door

Automatic Opening and Closing of Operator Door
Opening and closing of the operator door is automatically controlled to enable loading and unloading of the machine within an optimized cycle time. For short components (e.g. < 600 mm), the halves of the door can be decoupled from each other, thus making faster opening of one half possible and consequently an additional reduction in cycle time.

4.5.6 3-Axis Hybrid Laser Marker (Option)

Model		MD-X1000
Type		Marking unit (controller + head) 13W Standard area
2D code reader add-in (sold separately)		MD-XAD1/MD-XAD1A
Console (sold separately)		MC-P1
Marking method		XYZ 3-Axis simultaneous scanning method
Marking laser	Type	YVO_4 laser, Class 4 Laser Product (IEC60825-1, JIS C6802, FDA [CDRH] part 1040.10)*1
	Wavelength	1,064 nm
	Output	13 W
Q-switch frequency		CW (continuous wave), 1 to 400 kHz
Guide laser/Working distance pointer		Semiconductor laser, wavelength: 655 nm, output: 1.0 mW, Class 2 Laser Product (IEC60825-1, JIS C6802, FDA [CDRH] part 1040.10)*1
Marking area		125 x 125 x 42 mm

4.6 KUKA Standard Equipment Packages includes:

Painting

KUKA Deutschland offers the following single-colored paintings:

Machine housing	RAL 9003 signal white / RAL 7031 blue-grey
Control cabinet (incorporated)	RAL 9003 signal white
Robot/Logo	KUKA orange
Safety fence posts	RAL 7031 blue-grey
Safety fence grid	RAL 7031 blue-grey

KUKA Deutschland is prepared to offer multi-colored and special paintings against extra charge.

4.7 Project Management

At the start of the project, either party shall designate a project manager and a project team, as well as authorized signatories to carry out the agreed release and acceptance procedures. The project managers respectively designated by either party shall be authorized to make binding decisions concerning day-to-day business for the party concerned.

The parties will mutually arrange meetings for the planning and executing of the project.

Unless the contrary is expressly stated, the language of the project shall be German or English.

4.8 Engineering- and construction phase

The required performance data of the production system can only be assured if the relevant component and process information of the Contractor is made available to KUKA Deutschland.

KUKA Deutschland will carry out the construction and the assembly of the delivered system in accordance with the generally recognized rules of technology.

Notes on project execution

If any new findings concerning changed component geometry, tolerances or process technologies will be established during the execution of the project, the contract price will be adjusted in accordance with the new expenditure. If the Contractor fails to provide the machines, components and appliances in the tolerances/qualities required for the automatic production process in due time, KUKA Deutschland will notify the Contractor of the said omission and, after the expiration of a set grace period deliver the system in this design and in the quality achieved at that point in time. In this case, the Contractor is obliged to accept the system. A further optimization of the system by KUKA Deutschland at the Contractor's premises shall be invoiced only on a time and material basis.

Operating equipment instructions

The Contractor's operating equipment instructions shall not be taken into consideration, unless the contrary is expressly stated.

If KUKA Deutschland is requested to consider the operating equipment instructions or country-specific regulations after the placement of the order, additional charges and changes with regard to the delivery date may arise.

KUKA Deutschland uses the following tools:
- mechanical construction: CATIA V5 / NX10
- electrical construction: E-Plan P8, Output / pdf
- pneumatic plans: E-Plan P8, Output / pdf
-

4.9 Basic instruction and training

The scope of supply includes a basic instruction for the system. This shall be carried out during the commissioning phase. The Contractor is obligated to ensure the basic conditions required for operating and servicing the scope of supply, i.e. the operation of robots, systems and components. The Contractor shall guarantee that the selected personnel are highly experienced in handling the deployed technology.

The basic instruction consists of the following:
- 1 day for operators/setters
- ½ day for electrical maintenance and
- ½ day for mechanical maintenance

The basic instruction will take place during the day shift in German or English. Further languages can be offered on request. Besides the system documentation, no further training materials will be provided. If required, an extensive training can be offered.

The Contractor is obligated to deploy only properly trained and instructed personnel for operating the system.

4.10 Assembly, commissioning and programming

If the following works become a part of the scope of supply, KUKA Deutschland will undertake the installation of the system, starting from the distributor cabinet and/or sub-distribution main switch, and all necessary assembly work, including calibration by means of a laser tracker. The friction-welding machine will be transported to the installation site by the Contractor and set up in an oil tray. The Contractor shall install the main incoming supply with main and tap lines all the way to the feed points as specified in the installation plan.

The Contractor is obliged to provide the machines / components / appliances on time and in accordance with the tolerance values specified by KUKA Deutschland. This particularly applies to foundations as well as to interfaces on existing machines.

KUKA Deutschland assumes that the work will be performed without interruption; this applies to the assembly, commissioning and programming performed by KUKA Deutschland. If the Contractor is in delay with his performance, KUKA Deutschland reserves the right to charge the additional expenses arising out of the delay to the Contractor's account.

4.11 Pre-acceptance/delivery release at KUKA Deutschland

If a pre-acceptance/delivery release at KUKA Deutschland Augsburg was jointly agreed by the parties, KUKA Deutschland will notify the Contractor in writing of the readiness to perform the said pre-acceptance/ delivery release two weeks before starting the process.

The pre-acceptance/delivery release shall be carried out based on the following points:

- Visual check
- Proof of the specified features
- Confirmation of proper installation
- Accessibility for repair and maintenance work
- Ergonomics
- If agreed, cycle time in automatic operation
- Handover of the process parameters
- Welding of 50 components per acceptance component
- Check of the component tolerances acc. to KUKA producibility study*

Please note:

Single part drawings have to be provided by customer before order placement for creating of producibility study.

The acceptance component for pre-acceptance and final acceptance is defined before order placement. For setting the machine, the Customer will provide corresponding master samples and measuring devices free of charge. For test weldings the Customer will provide a sufficient quantity of 300 parts per acceptance component to KUKA Deutschland free of charge. These parts will be scrapped. Costs resulting from this will be charged to the Customer.

The components required for pre-acceptance will be made available to KUKA Deutschland free of charge in sufficient number and in the specified quality latest 6 weeks prior to pre-acceptance. Checks will be executed by the Customer and the results will be provided to KUKA Deutschland free of charge.

If the Customer cannot provide the components or fixtures in the required tolerances/qualities, KUKA Deutschland will make this a subject of discussion again and after a period of grace reasonable for the total situation will deliver the system with programs that will obtain the best possible quality. The Customer is obliged to accept the goods. A possibly required optimization of the programs at Customer's works by KUKA Deutschland in this case can only be invoiced according to expenditure.

In case the Customer requests support for parameter determination of additional component versions (or/and also testing of additional clamping tools), this expenditure will be invoiced in addition.

The costs for traveling and the hotel accommodation for participation in the pre-acceptance tests will be borne by the Customer. For repeated pre-acceptance, either party will bear its own costs.

In case pre-acceptance is postponed by more than two weeks or will not take place for reasons KUKA Deutschland is not responsible for, the acceptance is considered as effected with reference to the events triggering payment, and the readiness for shipment will come into force automatically. KUKA Deutschland is not responsible for resulting non-compliance of any agreed further dates. Due to such delays the Customer is not entitled to any claims for compensation of costs or damage or for agreed penalties. In return, KUKA Deutschland reserves the right to assert claims for compensation of the costs and damage resulting from such delays. Pre-acceptance at the Customer is not planned.

4.12 Final acceptance at the Contractor's premises

KUKA will unload the friction welding machine, transport to the installation site and pre-install it in the oil pan. A portable crane or other suitable transport system has to be provided by the customer.

The required media such as cooling water, air, and energy supply, etc. will be provided by the Customer free of charge at the positions shown in the layout. This work has to be done in due time before arrival of the KUKA Deutschland service technician.

KUKA Deutschland service technician will align the friction welding machine and check the mechanic and electric system as well as the functionality, and commissions the machine.

Immediately after commissioning the acceptance-test will be performed.

It consists of the following points:
- Proof of the specified features
- Handover of documentation
- Welding of 50 components per acceptance component
- Check of the component tolerances acc. to KUKA producibility study

The component type for the pre-acceptance and the final acceptance procedure are identical. Acceptance is documented by mutual signing of an acceptance report.
In any case, the system is considered as accepted latest one month after the planned final acceptance date if the final acceptance was postponed for reasons KUKA Deutschland is not responsible for or due to a mutual agreement. If no acceptance was agreed, the system is considered as accepted one month after delivery.
If the Customer's machines/ components/ devices do not maintain the required tolerance values, the Customer must immediately provide the corresponding remedy. Same applies for foundations, interfaces at existing machines etc. Extra costs incurred by this for KUKA Deutschland will be passed on to the Customer and will be invoiced separately.
Minor defects not hindering the intended use of the scope of supply will not obstruct the acceptance by the Customer, regardless of the obligation KUKA Deutschland has to eliminate these defects in a reasonable term. An acceptance with minor defects does not authorize the Customer to delay payment to be made on acceptance.
Approx. 2 weeks are provided as period for commissioning and acceptance (manually loaded machine) - for an automated cell approx. 2-3 weeks more have to be planned. The final acceptance will be performed based on the stand-alone friction welding machine (KUKA scope only). Final acceptance will be performed before any automation equipment will be installed.

The terms for assembly, commissioning and/or programming are based on a lump sum fixed price. As precondition, the Customer will have rendered all defined performances and preparations in due time. In case this is not fulfilled, any arising extra costs will have to be borne separately by the Customer.
In case assembly, commissioning and/or programming to be performed by KUKA Deutschland will be interrupted for reasons KUKA Deutschland is not responsible for, the Customer will have to bear extra costs possibly arising from waiting times, extra travels, extra charges etc.
KUKA Deutschland is not liable for costs, delays in the time schedule, and damage resulting from missing or delayed granting of official authorizations.

4.13 Documentation

According to the standards for final documentation of KUKA Deutschland and the valid standards and guidelines, the documentation shall be delivered in Dutch in triplicate, digital form. The documentation of subcontracted components will only be performed in original format and language.

- Installation drawing
- Circuit diagram
- Operating instructions
- Spare parts list
- Design drawings for wear parts in contact with the component
- Instructions for installation and commissioning
- Remarks concerning maintenance and repair
- Assembly drawings (pneumatics and mechanics)
- Safety instructions
- Machine lettering

4.14 CE Declaration of Conformity

EC Declaration of Conformity acc. to EC Machinery Directive 2006/42/EG
KUKA Deutschland complies with the relevant statutory provisions, standards and safety requirements. In case of deliveries to member states of the European Union, KUKA Deutschland will hand over a CE certificate.

4.15 Wear parts and maintenance

KUKA Deutschland' scope of supply includes wear parts only until the "finale acceptance" or "start of production", whichever event occurs first, is accomplished.

KUKA Deutschland will perform the maintenance intervals until the "finale acceptance" or "start of production", whichever event occurs first, milestone is achieved. After that, the Contractor has to carry out the maintenance work itself.

5 SERVICES PROVIDED BY CONTRACTOR

5.1 Provision of data by Contractor

The Contractor is obligated to hand over to KUKA Deutschland at the start of the project, but at the latest a week after placement of order, all the relevant data, giving the data inventory. The Contractor guarantees the correctness and completeness of the data provided.

In particular, KUKA Deutschland is not obligated to examine the technical data, drawings or any other information in the specifications of the Contractor and its suppliers, its specification sheet, its construction guidelines, nor any further documents that form an integral part of the contract or that are subsequently made available to KUKA Deutschland in the course of the project development.

In the event that the data, drawings or any other information of the Contractor should prove to be incorrect or incomplete at a later point in time or undergo a subsequent change, KUKA Deutschland will only carry out necessary changes to the construction and any further necessary measures only in return for the reimbursement of the extra costs that thereby arise.

CAD-Data

The Contractor is obliged to ensure that current, defect free CAD volume models, which comply with our current format specifications, are made available timely.
The quality of the CAD data of the production components must be usable without any additional work or preparation.
In addition to the CAD tolerances, all tolerance data for the components have also to be delivered. Any later corrections due to missing or defective tolerance data have to be processed by change management.

5.2 Project-specific free-issue parts of Contractor

In order to execute the offered scope of supply of KUKA Deutschland, the Contractor is obliged to provide the free-issue parts as listed below in the specified quality and quantity timely, accompanied by the required documentation in German, including the manufacturer's certificate, free of charge and ready for operation.

For the pre-acceptance

400 parts per component (quality acc. to KUKA Deutschland' feasibility study, measured with protocol)	8 weeks prior to pre-acceptance date

For final acceptance

400 parts per component (quality acc. to KUKA Deutschland' feasibility study, measured with protocol)	3 weeks prior to final acceptance date

Furthermore, the Contractor is obliged to perform the services required on site, timely.

5.3 Wear parts

The Contractor is obliged to provide wear parts for a production-capable system at an early stage, in order to avoid delays in the project schedule. A list of wear parts will be given to the Contractor after the end of construction.

5.4 Emission protection

In case emissions are generated during operation due to the scope of supply, the scope of supply may only be operated by the operating company using an equipment (e.g. exhaust system) meeting the valid directives and legal emission limit values.

5.5 Noise protection

Unless listed in the scope of supply of KUKA Deutschland, the Contractor is obliged to ensure that noise protection measures and secondary measures, particularly protective walls or enclosures/booths, are warranted.

6 TECHNICAL CONDITIONS

6.1 Technical data

Based on the data handed over by the Contractor, KUKA Deutschland made the following assumptions:

Forging force*	12 to 300 kN
Spindle speed (rpm)**	3,000 min-1 max.
- with hydraulic brake**	2,200 min-1 max.
- when using of EPE**	1,800 min-1 max.
Slide stroke*	395 mm max.
Component length (spindle side)*	300 mm max. (with Ø 65 mm)
Component length (slide side)*	1200 mm max. (or on request)
Component diameter (slide side)*	100 mm max. (or on request)
Weight*	approx. 12,000 kg
Dimensions (L x W x H)*	approx. 5,500 x 2,100 x 2,400 mm
Noise pressure level*	approx. 78 dB (A)
Installation altitude	1,000 m above sea level max.
Ambient temperature	5°C – 40 °C
Humidity	90 % (max. 20 °C)
	50 % (max. 40 °C)
Electric (Mid-range-drive)	
Supply voltage	400 V +/- 10%
Total connected power	145 kVA
Average power consumption*	90 kW
Cable cross section	150 mm²
Main fuse	400 A (slow acting)
Cooling water	
Inlet temperature	20 - 25°C
Supply pressure	3 - 6 bar
Δ Pressure	≥ 2.5 bar
Cooling water requirement	approx. 20 l/min
pH value	7.5 - 8.5
Particle size of impurities	≤ 0.15 mm
Conductivity	≤ 500 µS/cm
Outlet temperature	40°C max.

*) Depending on welding process and component-specific machine design
**) Max. spindle speed is dependent on the forge force used

Based on the information handed over by the Contractor, as well as the exclusions under "Cycle time" and "Technical availability", the offered system has the performance data as specified above.

6.2 Welding quality

In order to guarantee a continuous quality of welding parts and a smooth welding process, it will be necessary to agree on the quality of the individual components. The quality of individual components shall be jointly agreed upon by the parties, in accordance with the process-related and geometrical requirements.

The single part quality must be agreed between KUKA and the Customer according to the process technical and geometric requirements.

The feasibility of the geometric, mechanic, physical and metallurgic features of the weld part is checked on the basis of measuring results, experience, single part quality and the specifications made available by the Customer. Values of the geometrical features that can be reached are documented for the corresponding weld part for the producibility study to be established.

The necessity of welding tests in order to verify calculated figures will be determined by the responsible employees of the corresponding departments.

Component-specific examinations and the validation of the components shall be performed and/or arranged by the Contractor, and the resulting parameters and results will be provided in written form and free of charge to KUKA Deutschland for further tests and/or release for series production. The costs of the tests are not considered in the indication.

7 BASIC REQUIREMENTS REGARDING THE INSTALLATION SITE OF A SYSTEM

7.1 Quality / structure

The following basic requirements to quality and structure have to be met for the installation site of a system:

- Concrete quality B25 according to DIN 1045 (1988) or C20 / C25 according to DIN EN 206/1 (2001)
- Compound screed (industrial design) is admissible (some centimeters thick, max. 5 cm).
- Screed on a separation layer is not permitted!
- Screed on an insulation layer is not permitted!
- Minimum thickness of floor and ceiling plate must be min. 225 mm (for machine and robot fixation)

7.2 Separating and expansion joints, minimum spacings

Individual main components should be standing on one consistent base plate. If this cannot be put into practice, please consult the manufacturer. A separating or expansion joint between main components may have negative impact on the process concerned.

The distance c > 25 cm in all directions between separating and expansion joints, concrete edges and bore holes generally must not be fallen below.

7.3 Surface / Accuracy of flatness

Largely even floor, without stairs and edges. The maximum height difference (highest point to deepest point) must not exceed 10 mm in the complete installation area (for big systems referred to system areas).
For the installation area of individual main components, e. g. industrial robot or robot platform, locally higher requirements are made to evenness.

A superimposed, comprehensive examination for the complete building with regard to the static and dynamic loads by the Customer is mandatory and is not in the responsibility of KUKA Industries.

7.4 Other boundary conditions

Besides this, vibrations (originating from heavy machines and systems, e. g. presses) or variable loads (e. g. travel paths with heavy fork lift truck operation) have to be considered. In case of doubt, the individual case must be considered.

7.5 Static equilibrium and building

According to experience, only the static surface load kN/m² is known for buildings. The complete building and the installation site must comply with the criteria defined in the above chapter. Due to the installed main components (e. g. machine, robot, but also fork lift trucks, cranes, etc.), besides the static loads, also high dynamic (complex) loads are induced in the building. Weak points (slur of distributed load) and the simultaneity of loads may also be considered. Due to these forces and moments the static equilibrium of the building must not be exceeded. The issue "separation of reinforcing steel" must be considered with examinations and calculations. For this purpose a practice-oriented approach must be defined. Therefore a superimposed load examination and statics calculation for the complete building by the Customer is mandatory.

8 PREPARATION OF COMPONENTS AND WELDING QUALITY

To ensure a faultless welding cycle and to keep specified component tolerances, the Customer must maintain and guarantee a constant quality of the components and of the preparation of components.

Prior to placement of order, the component and the preparation of components has to be defined and fixed.

The bead height and shape as well as weld spatters are depending on the component's preparation, batch, and so on. Therefore the bead shape and height as well as total absence of spatters cannot be guaranteed.

The weld surfaces must be metallic bright and free from dirt, lubricants and cooling agents.

The components have to be processed at both sides at the stop and weld surfaces in rectangular manner. A maximum wobble of 0,5 mm over the complete cross section and less than 0,5°, referred to both, has to be observed.

Compliance with defined component tolerances depends on the component preparation, the material, the component geometry and on the clamping and reference surfaces. A statement on tolerance values, mechanical, physical and metallurgical features is only possible after execution of weld trials. A feasibility study without previous weld trials is based on experience values and in case of doubt is not relevant for acceptance. The weld parameters, material allowance, tolerances before and after welding, heat treatment, etc. are empirically determined in the trials and finally the preparation of the components and the weld cycle as well as acceptance criteria for the serial friction welding machine are defined.

Component-specific tests and the validation of the components have to be executed by the Ordering Party and the parameters and findings resulting for further trials and / or release for mass production must be approved in writing and made available to KUKA free of charge. The costs for the trials are not included in the indication.

The results gained from the trials and fixed definitions regarding preparation of components, tolerances, clamping and reference surfaces, material, pre- and post-heat treatment, hardness, weld time, material allowance, weld parameters, etc. are the basis for acceptance of the machine.

The machine parameters are controlled and monitored during the weld process by the machine control. The Ordering Party is responsible for compliance with component influences and the weld cycle and thus for the welding quality after acceptance/ start of mass production.

9 TOLERANCE CRITERIA

Dimensional deviations of the welded parts are influenced by the weld process and the admissible dimensional deviations of the single parts. The tolerances of the individual parts in total must be less than the tolerances of the welded parts (reference value 50 %).

The quality-relevant features of the individual parts have to be measured and documented prior to acceptance by the Customer.

For the individual parts at least the same quality criteria must apply as for the welded ones, i. e. for the relevant features of the individual parts the Customer must perform a capability proof (Cm; Cmk ≥ 1,33).

Acceptance as well as statistic studies (if required) are at the Customer's charge.

The applicability of the statistic evaluation must be agreed in writing between KUKA Industries and the Customer.

Possible tolerances must be discussed and fixed in a separate discussion between the Customer and KUKA Industries´ special departments process technology and design. For this purpose a "feasibility study" is established, taking into consideration the tolerances of the individual parts.

10 PRICE BREAKDOWN

Prices do not include VAT. Prices are given in EURO.

10.1 Scope of Supply

Qty.	Description	Total Price [EUR]
1	Friction Welding Machine, Genius plus	incl.
1	Project management	incl.
1	Change management	incl.
1	Pre-Acceptance at KUKA Augsburg	incl.
1	Assembly, commissioning, and final acceptance at the Customer	incl.
1	Documentation in English	incl.
1	CE Declaration of Conformity	incl.
1	Freight and packaging, FOB Hamburg Harbour	incl.
1	Exhaust hole in machine covering	incl.
	Price:	1,097.270 €

10.2 Price of Options

Qty.	Description	Total Price [EUR]
1	Flash removal package (Position 4.3.1	62.500 €
1	Tool Changer (Position 4.3.2)	8.250 €
1	Steady Rest (Position 4.3.3)	5.650 €
1	Chip conveyor (Position 4.3.4)	12.050 €
1	Automatic Door (Position 4.3.5)	3.350 €
1	Keyence MD-X1000 (Position 4.5.6)	7.280 €
	Express Delivery by Air (instead of Ship/Vessel)	78.000 €

11 COMMERCIAL TERMS

11.1 Payment Terms

The price shown above does not include any duties or taxes. Payment terms shall be as follows:

30 % upon placement of an order by T/T, payable within 30 days after invoicing
30 % after design freeze by T/T, payable within 30 days after invoicing
40 % by irrevocable L/C to be opened by a first class bank within 6 weeks from placement of order, allowing partial shipment, according to the following payment plan:

30 % of total contract value payable at sight against presentation of commercial invoice, packing list, transportation documents.

10 % of total contract value payable at sight against presentation of final acceptance certificate, however not later than 3 months after the shipping date of the last partial shipment, even after expiry date of L/C and even without presentation of final acceptance certificate. Presentation of final acceptance certificate after expiry date of L/C is also acceptable.

The L/C is subject to the Uniform Customs and Practice for Documentary Credits, ICC Publication No. 600 and is to be advised through, available with and confirmed by Commerzbank AG, Swift/ BIC-Code COBADEFF.

11.2 Partial delivery / partial invoicing

If the order consists of several sub-projects or jobs that must be completed independently and that will, if necessary, be delivered on different dates, partial deliveries and partial invoicing are possible.

11.3 Time Schedule

Approx. 11-12 months after order and agreement on all technical and commercial terms until pre-acceptance run.

11.4 Terms of delivery

KUKA Deutschland undertakes to deliver the Contractual Goods FCA Augsburg (acc. to Incoterms 2020).

In case of deliveries to other EU countries, the Contractor is obliged to confirm the receipt of the Contractual Goods by sending a written acknowledgement of receipt to KUKA Deutschland, in conformity with the German tax law provisions.

Acceptance by national authorities

Acceptance by national authorities is not included in KUKA Deutschland' scope of supply. If this is requested, all costs, including those arising due to requests for changes from the authority, shall be borne by the Contractor.

11.5 Change Management

The term "change" refers to any subsequent modification of the agreed scope of supply.

Changes could affect the originally agreed quotation price and the time schedule. KUKA Deutschland reserves the right to examine the changes requested by the Contractor and to inform the Contractor of their potential impact.

Changes to the agreed scope of supply shall be presented by KUKA Deutschland in a written supplementary quotation. The precondition for making such changes is that all requested modifications are mutually and fully agreed upon by the parties, and that KUKA Deutschland receives a written order which fully complies with the content of the quotation.

12 LEGAL CONDITIONS

12.1 Warranty and Liability

Unless otherwise expressly agreed, any warranty or liability of KUKA Deutschland for defects of the Contractual Goods shall be defined by applicable law.

The warranty period with reference to the scope of supply of the quotation will be 24 months. In case of quotations for work and services the warranty period shall begin with start of production (SOP) or scheduled final acceptance, whichever comes first. In case of purchase and other delivery contracts without final acceptance the warranty period shall begin from the passing of risk according to the agreed terms of delivery.
KUKA Deutschland will not give any warranty and shall not incur any liability in connection with the use of free-issue equipment and the reuse of any existing devices, components or line parts as well as for components resulting from tests or from the production of prototypes.

KUKA Deutschland shall be liable, whether for breach of contract, by way of an indemnity, in tort or on any other legal grounds whatsoever, only for gross negligence or willful conduct, unless in cases of (i) death or personal injury, (ii) breach of an obligation essential for this type of contract or (iii) liability of KUKA Deutschland to Contractor which may not be limited or excluded according to mandatory provisions of law. In case of breach of an essential contractual obligation due to simple negligence, KUKA Deutschland however shall only be liable for damages reasonably foreseeable.

Furthermore any liability of KUKA Deutschland, whether (i) based on breach of contract, (ii) by way of an indemnity, (iii) in tort or (iv) on any other legal grounds whatsoever, for any consequential and / or indirect damages (in particular, but not limited to, lost profits, lost revenues, lost opportunities and damages due to business interruption), is excluded, unless such damages have been caused willfully. For the avoidance of doubt, the liability of KUKA Deutschland for lost profits, lost revenues, lost opportunities and damages due to business interruption shall be excluded also, if such damages are qualified as direct damages according to applicable law.

To the extent that KUKA Deutschland' liability is excluded or limited on the basis of the foregoing provisions, the exclusion or limitation shall apply accordingly for legal representatives, mediators and vicarious agents of KUKA Deutschland.

12.2 Reservation of Title

KUKA Deutschland retains title to the scope of supply until receipt of all payments and fulfillment of all other claims against the Contractor arising from the order.

12.3 Force Majeure

Delays or the failure of the performance within the framework of the contract as a result of an event of force majeure without error or fault on the part of the Contractual Partner concerned shall be deemed excused as long as the event continues. This presupposes that the affected Contractual Partner informs the other Contractual Partner in writing immediately after the occurrence of the event of force majeure, but no later than 3 days thereafter, of the type and extent of the event of force majeure that has occurred and its effects, including the probable duration.

Events of force majeure are unforeseeable, unavoidable, and extraordinary events such as natural catastrophes such as floods, earthquakes, hurricanes or other extreme natural events, shortages of raw materials, energy, and labor, industrial disputes, involuntary or unforeseeable operational disruptions, fires, unrest, wars, sabotage, terrorist attacks as well as the outbreak of an epidemic or pandemic.

If the affected Contractual Partner cannot provide credible assurance that a delay due to force majeure will not exceed 60 days or if a delay due to force majeure exceeds 60 days, the other party may terminate the contract without liability.

If, due to the current pandemic (coronavirus), the Russia/Ukraine war or due to delivery difficulties of components or raw materials and the consequences thereof, the execution of the agreement is delayed in whole or in part or becomes impossible (in particular shortage of raw materials or components, entry bans, shutdown), neither of the parties shall be responsible and the resulting liability of the parties is excluded (including contractual penalties, interest on arrears, damages caused by delay). If one of the parties is affected by a delay/impossibility, it shall inform the other party thereof, stating the respective event and the expected duration. Any delivery or performance deadlines - even during the period of delay - shall be extended by the duration of the disruptions in performance caused by these circumstances; this shall also apply if such circumstances occur at subcontractors. In case the above mentioned circumstances, directly or indirectly, result in increased costs of Supplier's contractual performance, the parties shall agree in good faith upon an equitable adjustment of the contract price.

In the event of impediment to performance, over a period of more than 4 months or a repeated significant impediment to performance, the parties shall negotiate an adjustment of the contract upon written request by one of the parties.

12.4 Export control

The supplies and performances (performance of contract) are subject to the reserve that the fulfilment is not prevented by any impediments arising out of national or international regulations, especially export control regulations and embargoes or other sanctions. The parties undertake to provide all information and documents required for export / transfer / import. Delays due to export examinations or approval procedures will invalidate terms and deadlines. If required authorizations are not granted, the contract regarding the parts concerned shall be considered as not concluded; claims for damages will be excluded in so far and due to missed deadlines mentioned above.

12.5 Industrial Property Rights

KUKA Deutschland grants to the Contractor a non-exclusive and non-transferable right of utilization to all the findings and inventions embodied in the scope of supply - no matter if they have been (i) already found before start of the order or only developed in the course of it and if they are (ii) patentable or not - for the purpose of start-up, operation, maintenance and repair of the scope of supply under this contract. The transfer of further rights is subject to a separate written agreement. KUKA Deutschland reserves all rights of any kind for provided documents and information, especially any rights to inventions and (industrial) property rights.

12.6 Compliance

The parties declare their commitment to a corruption-free business community. They under-take to refrain from a) any kind of corrupt conduct and b) any other kind of criminal practices, and to take all necessary measures to prevent such conduct.

In the event of a breach of this obligation by either party, the other party is entitled to terminate any contract for this project without prior notice.

Furthermore, in the event of a breach of this obligation by either party, the other party is entitled to discontinue any business relationships with the breaching party without incurring any liability to the breaching party.

12.7 General Provisions

KUKA Deutschland' quotation represents the exclusive basis of the contract to be concluded with an order from the Contractor. Any terms and conditions of the Contractor which deviate from the terms and conditions of KUKA Deutschland' quotation are not accepted. The fact that KUKA Deutschland renders services or accepts payments without expressly raising any objections against such terms is not to be interpreted as an acceptance of the Contractor's terms and conditions, and under no circumstances constitutes conclusive conduct implying intent to conclude a contract under the conditions of the Contractor.

The formation of a contract, its validity, termination, interpretation, execution and the settlement of any dispute hereunder shall be governed by the Laws of Germany, excluding its Conflict of Laws provisions and the provisions of the United Nations Convention on Contracts for the International Sale of Goods (CISG). The courts of Augsburg (Germany) shall have exclusive jurisdiction.

13 CUSTOMER SERVICE

KUKA Deutschland offers an extensive range of consulting and other services, ranging from staff training at the KUKA Colleges and the 24/7 hotline, to the modernisation and refurbishment of robots, systems, complete systems and the delivery of spare parts, on-site service and the sale of used machines. Enquiries regarding these services should be addressed to KUKA Deutschland' Customer Service **Augsburg (consulting.cs.Deutschland.de@kuka.com // +49 821 797-1001).**

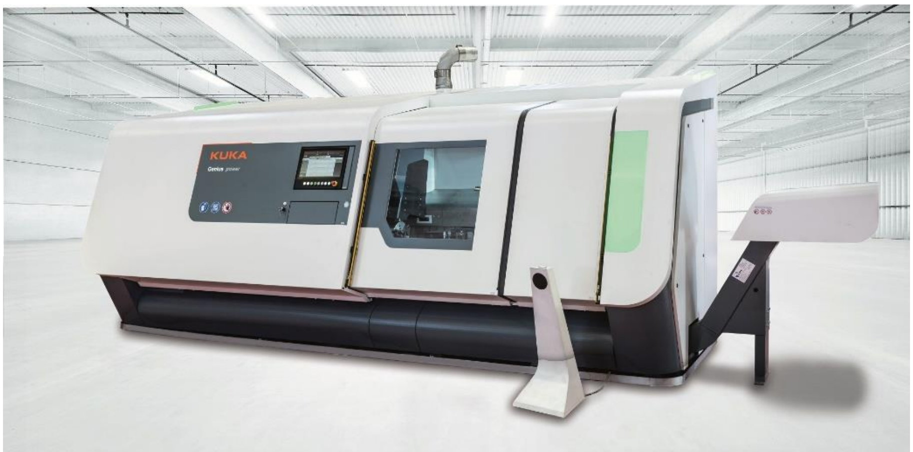
There are already existing contracts in place for customer service with the previous machine. KUKA suggests to refer to these contracts and rates related, for more details see **오류! 참조 원본을 찾을 수 없습니다.**.

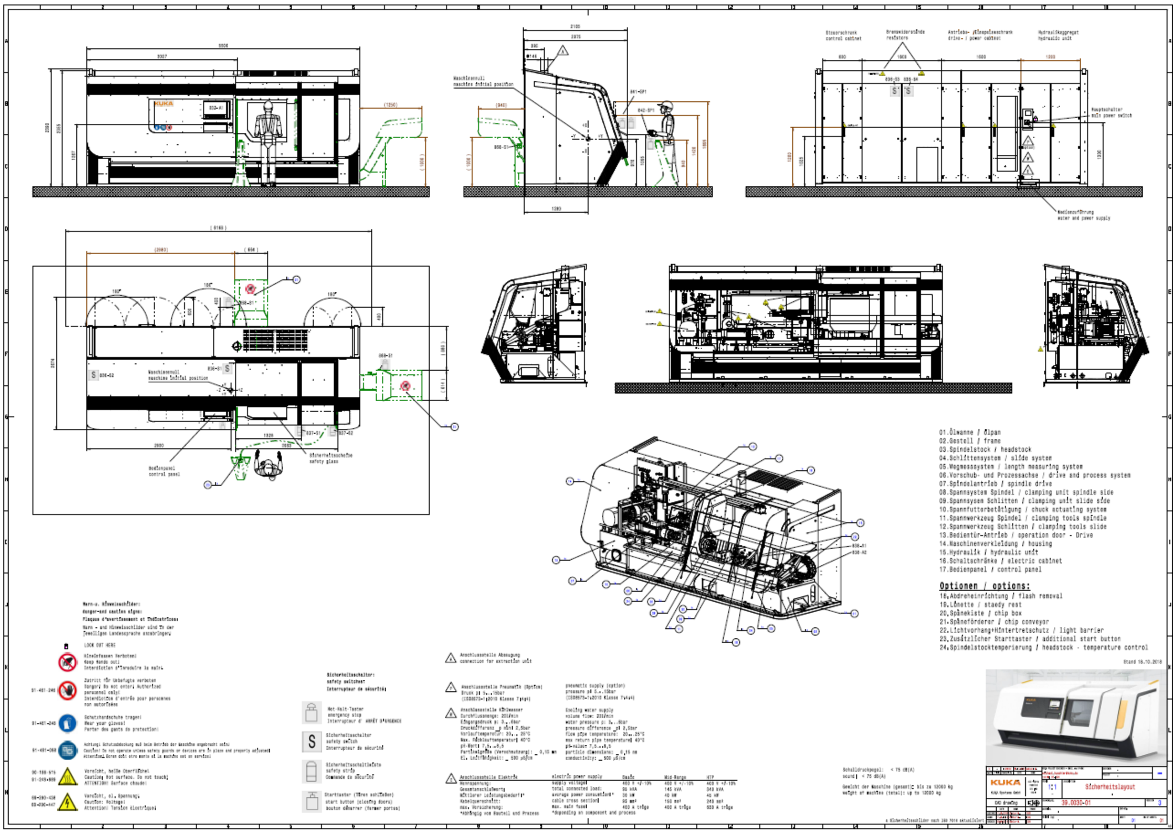
Weitere Informationen sind unter nachfolgendem Link erhältlich: https://www.kuka.com/de-de/services.

14 VALIDITY OF QUOTATION

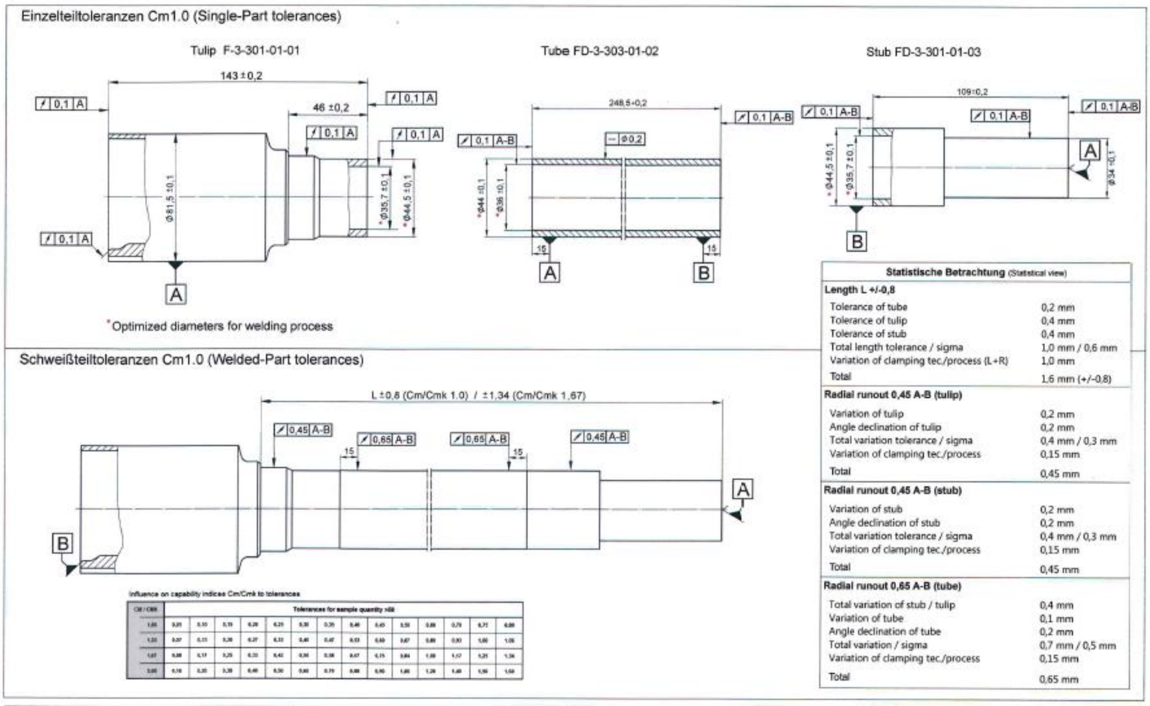
This quotation is valid until 31.12.2022

15 LAYOUT / PICTURES





16 WELDING STUDY



Machbarkeitsstudie (Producibility-Study)	Version	Bauteilbezeichnung (Part description)	Zeichnungsnummer (Drawing-No.)	Revision	Team
	02	Tulip welded	FD-3-303-01	03	Prozesstechnik (Process technology): Klett, Ohm
		Schweißverfahren (Welding process): Friction-welding	Maschinentyp (Machine-type): Genius	Werk-Nr. (Serial-no.): NA	Qualitätssicherung (Quality departement): Kastl; Konstruktion (Design departement): Büchler

KUKA Systems GmbH

Einzelteiltoleranzen Cm1.0 (Single-Part tolerances)

*Optimized diameters for welding process

Schweißteiltoleranzen Cm1.0 (Welded-Part tolerances)

Influence on capability indices Cm/Cmk to tolerances

Cm / Cmk	Tolerances for sample quantity >50													
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Statistische Betrachtung (Statistical view)	
Length L +/-0,8	
Tolerance of tube	0,2 mm
Tolerance of tulip	0,4 mm
Tolerance of stub	0,4 mm
Total length tolerance / sigma	1,0 mm / 0,6 mm
Variation of clamping tec./process (L+R)	1,0 mm
Total	1,6 mm (+/-0,8)
Radial runout 0,45 A-B (tulip)	
Variation of tulip	0,2 mm
Angle declination of tulip	0,2 mm
Total variation tolerance / sigma	0,4 mm / 0,3 mm
Variation of clamping tec./process	0,15 mm
Total	0,45 mm
Radial runout 0,45 A-B (stub)	
Variation of stub	0,2 mm
Angle declination of stub	0,2 mm
Total variation tolerance / sigma	0,4 mm / 0,3 mm
Variation of clamping tec./process	0,15 mm
Total	0,45 mm
Radial runout 0,65 A-B (tube)	
Total variation of stub / tulip	0,4 mm
Variation of tube	0,1 mm
Angle declination of tube	0,2 mm
Total variation / sigma	0,7 mm / 0,5 mm
Variation of clamping tec./process	0,15 mm
Total	0,65 mm

Erstellt (Created): OQ-A Kastl 2016-01-25 | Geändert (Changed): | Seite (page) 1 von (of) 1